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Self-rated oral health, but not tooth loss, is associated with postpartum depressive symptoms: a nationally representative cross-sectional study with a pre-registered negative control

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What this paper adds

What is known.

Poor oral health and depression are associated in adults, and gum problems in pregnancy are associated with depression in pregnancy. Prospective cohorts in older adults report that tooth loss precedes depressive symptoms.

What this study adds.

In a nationally representative sample of US women 1–24 months after a birth, poorer self-rated oral health was strongly associated with depressive symptoms, whereas the clinically counted number of missing permanent teeth — an exposure that accumulates over years and therefore cannot have been produced by depression arising after the birth — was not associated at all. A pre-registered negative control (missing third molars) was also null.

Implication.

In this population the oral health–depression association is carried by current, self-rated oral state rather than by accumulated oral disease. It is therefore unlikely to represent a causal effect of oral disease on postpartum depressive symptoms, and dental measures aimed at preventing postpartum depression should not be justified on this evidence.

Abstract

Objectives.

Poor oral health is associated with depression, but the direction of the association is unresolved and the postpartum period is largely unstudied. We tested whether the association is carried by accumulated oral disease, whose timing precedes a recent birth, or by current self-rated oral state.

Methods.

Cross-sectional analysis of the US National Health and Nutrition Examination Survey (NHANES), 2007–March 2020, comprising 861 women 1–24 months after a birth (742 with a dentition examination). Depressive symptoms were measured with the PHQ-9 (total score; ≥ 10 as screen positive). Exposures were self-rated health of teeth and gums (5-point scale) and the examiner-counted number of missing permanent teeth excluding third molars. Missing third molars, usually extracted electively, served as a pre-registered negative control. Design-based linear and logistic regression (Taylor linearisation) adjusted for age, race/ethnicity, education, income-to-poverty ratio, nativity, serum cotinine, body mass index and survey period. Hypotheses, models and numbered predictions were registered before analysis.

Results.

Each one-step worsening of self-rated oral health was associated with a 0.68-point higher PHQ-9 score (95% CI 0.27–1.08; $p = 0.001$) and with screen positivity (OR 1.98, 95% CI 1.38–2.83; $p < 0.001$). Each additional missing permanent tooth was associated with a -0.02 -point change (95% CI -0.14 to 0.09 ; $p = 0.68$); the standardised estimate excluded associations above 0.09 SD. The negative control was null ($p = 0.39$). With both exposures in one model, self-rated oral health retained its association (0.71 ; 95% CI 0.30 – 1.11) and tooth loss remained null (-0.11 ; 95% CI -0.27 to 0.06). Tooth loss did predict poorer self-rated oral health (0.08 per tooth; $p < 0.001$).

Conclusions.

Accumulated tooth loss was unrelated to postpartum depressive symptoms, while concurrent self-rated oral health was strongly related. The association is more consistent with reverse causation or shared current state than with a causal effect of oral disease.

1. Introduction

Depression after childbirth affects roughly one in eight women and carries consequences for mothers, infants and families.¹ Identifying modifiable causes matters clinically, but most candidate factors are known only through cross-sectional associations, and umbrella reviews of postpartum depression risk factors rate the certainty of most associations as low or very low.¹

Oral health is one such candidate. Poor oral health and depression co-occur in general adult populations, and plausible pathways run in both directions: chronic oral pain and periodontal inflammation could contribute to depressive symptoms, while depression reduces oral self-care, diet quality and dental attendance.^{2–4} In pregnancy, self-reported gum problems were associated with depression in a database of 316,956 deliveries,⁵ and depression was associated with poorer oral health among pregnant women in Appalachia.⁶ A 2025 rapid review of oral health and psychological status from pregnancy to one year postpartum concluded that these relationships remain understudied, particularly after delivery.⁷ Longitudinally, depression and stress predict subsequent decline in oral self-care in perinatal women.⁸

Evidence that oral disease precedes depression comes mainly from older adults: in a 20-year Japanese cohort, fewer remaining teeth predicted incident depressive symptoms;⁹ a rural Chilean cohort found higher odds of incident depression with fewer teeth;¹⁰ a Taiwanese study found that losing teeth between examinations predicted later psychological distress;¹¹ and in the English Longitudinal Study of Ageing, poorer oral-health-related quality of life predicted later depressive symptoms.¹² Genetic (Mendelian randomization) studies disagree: one supports a bidirectional oral health–psychiatric link,¹³ one supports depression → oral disease,¹⁴ and two find no causal relationship, one of them after adjustment for smoking and education.^{15,16}

The postpartum period is a specific and under-examined context. Pregnancy worsens gingival health, dental care is frequently deferred, and depressive symptoms are common, so an association is expected whatever its direction. The central difficulty is that self-rated oral health and depressive symptoms measured at the same moment cannot be ordered in time, and self-rating may itself be coloured by mood.

We used a feature of the dentition to address this. Teeth do not regenerate: the number of missing permanent teeth is the cumulative record of decay and periodontal disease over years, so all but a small part of it necessarily predates a birth in the previous 1–24 months. If accumulated oral disease contributes causally to postpartum depressive symptoms, tooth loss should be associated with them. If instead the association reflects current perception or reverse causation, self-rated oral health should be associated while tooth loss is not. We pre-registered this contrast, together with a negative control — missing third molars, usually removed electively rather than lost to disease — and analysed a nationally representative sample.

2. Methods

2.1 Design and data source

We analysed the US National Health and Nutrition Examination Survey (NHANES), a continuous, nationally representative survey of the civilian non-institutionalised population combining a household interview with a standardised examination in a mobile examination centre.¹⁷ We pooled six cycles: 2007–08, 2009–10, 2011–12, 2013–14, 2015–16 and the pre-pandemic 2017–March 2020 file (13.2 survey-years). The 2017–18 cycle was not added separately because the pre-pandemic file already contains those respondents. Cycles before 2005 lack the PHQ-9, and 2005–06 and 2021–23 lack the months-since-birth item.

Public-use demographic, depression-screener, reproductive-health, dentition, oral-health-questionnaire, body-measures, biochemistry, complete-blood-count, cotinine and urinary-creatinine files were merged on the survey participant identifier within NHANES only. Reporting follows STROBE for cross-sectional studies.¹⁸

2.2 Study population

Eligible women reported a birth 1–24 months previously (item RHQ197), were not currently pregnant by self-report or examination pregnancy test, had a positive examination weight, and had a complete PHQ-9 ($n = 861$). Analyses of tooth counts were restricted to those with a dentition examination ($n = 742$). The 1–24 month window was pre-registered as the primary analysis pool in earlier work on this cohort because it roughly doubles the sample available in the 1–12 month window.

2.3 Outcome

Depressive symptoms were measured with the nine-item Patient Health Questionnaire (PHQ-9), administered in the examination centre, scored 0–3 per item (range 0–27), and required complete items.¹⁹ We analysed the total score and the conventional screening threshold of ≥ 10 . Internal consistency in the analytic sample was good (Cronbach's α 0.84, 90% CI 0.83–0.85). A previous NHANES analysis found the PHQ-9 measurement-invariant across pregnant, postpartum and non-peripartum women, with only clinically negligible item differences.²⁰ A score of ≥ 10 indicates screening positivity, not a clinical diagnosis.

2.4 Exposures

Self-rated oral health. “How would you rate the health of your teeth and gums?”, answered excellent, very good, good, fair or poor, and analysed as a 1–5 score with higher values worse (asked from 2009–10 onward).

Tooth loss (primary, timing-fixed exposure). In the dentition examination a trained examiner records the status of each of 32 tooth positions. We counted positions recorded as “tooth not present” among the 28 non-third-molar permanent teeth. The count was computed only when every tooth in the group was assessed and at least 20 positions were assessed overall, so a partial examination is missing data rather than a spuriously small count. Per-tooth codes are identical in the 2007–08 file and in the 2009-onward files.

Negative control. The same count over the four third molars. Third molars are commonly extracted electively in adolescence, on orthodontic advice and with dental access, rather than lost to untreated disease; they therefore share the socioeconomic and dental-contact structure of other tooth loss but not its disease, pain or disfigurement.²¹

2.5 Covariates

Covariates were fixed before analysis as common causes of oral status and depressive symptoms: age (linear and quadratic), race/ethnicity, education, family income-to-poverty ratio, nativity (born in the US), serum cotinine (tobacco exposure), body mass index and survey period. Self-rated oral health was pre-specified as a *mediator* of tooth loss and was therefore not included in the primary tooth-loss model.

2.6 Statistical analysis

All estimates used the pooled examination weights, with strata and primary sampling units unique to each cycle, and Taylor-linearisation variance estimation (design $df = 104$).²² Subpopulations were analysed by zero-weighting and never by deleting rows, as the NHANES analytic guidelines require.²² Linear models were used for the PHQ-9 total and logistic models for the ≥ 10 threshold.

Pre-specified analyses were:

- adjusted models for each exposure and each outcome;
- dose-response across pre-specified tooth-loss categories (0, 1–2, 3–5, 6+) and across the five oral-health ratings;
- the negative-control model;
- a joint model containing self-rated oral health and tooth loss together;
- the first step of the proposed pathway, tooth loss $\rightarrow$ self-rated oral health;
- the same models in a comparison group of women aged 20–44 who were not pregnant and had had no birth in 24 months;

- a model restricted to the five cognitive-affective PHQ-9 items, excluding the sleep, fatigue, appetite and psychomotor items that infant care disturbs;
- sensitivity of any positive finding to unmeasured confounding, using the robustness value and omitted-variable bias bounds benchmarked on the strongest measured confounder.²³

Analyses were complete-case; the number contributing to each model is reported. Because this analysis belongs to a larger pre-registered search programme on the same respondents, false-discovery control (Benjamini–Hochberg)²⁴ was applied across every hypothesis tested in that programme (68 hypotheses), not only the hypotheses reported here.

2.7 Pre-registration, code and reproducibility

The analysis plan, exposures, covariates, categories, the negative control and four numbered predictions were registered in the study repository and committed before the analysis was run; the predictions included that tooth loss would be associated with symptoms and that the negative control would not. Analysis code, the pre-registration and all generated tables are in the repository (Supplementary Files 1–4). No analytic decision was changed after results were seen, and the prediction that failed is reported as such.

2.8 Ethics and participant privacy

NHANES protocols were approved by the NCHS Research Ethics Review Board and all participants gave written informed consent. This secondary analysis of de-identified public-use data required no further approval. No individual-level data were exported. Every published figure is a weighted aggregate; no count, percentage or mean resting on fewer than five women (or fewer than five cases or non-cases) is shown, and no minimum or maximum values are reported.

3. Results

3.1 Sample

Of 861 women 1–24 months postpartum with a complete PHQ-9, 742 (86%) had a dentition examination (Table 1). Mean age was 29.7 years, mean months since birth 11.0, 80% were married or living with a partner, and 23.8% had serum cotinine ≥ 3 ng/mL. Mean PHQ-9 was 3.33 and 8.7% screened positive. Two-thirds (67.2%) had no missing permanent teeth, 16.4% had 1–2 missing, 12.8% had 3–5 and 3.7% had six or more; 61.5% were missing all four third molars. Self-rated oral health was fair or poor for 22.5% of the analytic sample.

3.2 Self-rated oral health

Mean PHQ-9 rose monotonically across oral-health ratings, from 2.15 (95% CI 1.38–2.92) among women rating their teeth and gums excellent to 6.41 (4.70–8.11) among those rating them poor; screen positivity rose from a suppressed cell (<5 cases) to 27.6% (16.8–41.9) (Table 2, Figure 1). In adjusted models, each one-step worsening was associated with a 0.68-point higher PHQ-9 total (95% CI 0.27–1.08; $p = 0.001$; $n = 577$) and with higher odds of screening positive (OR 1.98, 95% CI 1.38–2.83; $p < 0.001$) (Table 3).

The association survived the pre-specified robustness checks: it replicated in independent halves of the survey strata (disattenuated correlations +0.254 and +0.177), held when restricted to the five cognitive-affective PHQ-9 items (+0.290; $p = 0.018$), and appeared with the same gradient in women with no recent birth. Somatic items (sleep, fatigue, appetite, psychomotor) carried 58% of the association against an item share of 44%, below our pre-set flag of +15 percentage points. An unmeasured confounder would have to explain 11.9% of the residual variance of both exposure and outcome to render it non-significant, and 27.5% to reduce it to zero.

3.3 Tooth loss and the negative control

Tooth loss showed no association with depressive symptoms (Table 3). Each additional missing permanent tooth was associated with a –0.02-point change in PHQ-9 (95% CI –0.14 to 0.09; $p = 0.68$; $n = 658$) and with an odds ratio of 1.01 for screening positive (95% CI 0.94–1.09; $p = 0.79$). On a standardised scale the estimate was –0.015 SD (95% CI –0.087 to 0.057), excluding associations larger than about 0.09 SD — smaller than the association observed for self-rated oral health

in the same women. There was no gradient across categories: mean PHQ-9 was 3.31, 3.60, 3.22 and 2.83 for 0, 1–2, 3–5 and 6+ missing teeth (Table 2).

The pre-registered negative control was likewise null: each missing third molar was associated with a 0.11-point change in PHQ-9 (95% CI –0.14 to 0.36; $p = 0.39$), so the null for tooth loss cannot be attributed to a suppressive artefact of elective dental care.

In the comparison group of 4,458 women aged 20–44 without a recent birth, tooth loss was also unassociated with the PHQ-9 total (0.04 points per tooth; 95% CI –0.02 to 0.10; $p = 0.23$), which rules out low power in the postpartum sample as the sole explanation.

3.4 The two exposures together

Tooth loss predicted self-rated oral health as expected: each missing tooth was associated with a 0.08-point worse rating (95% CI 0.03–0.13; $p < 0.001$; $n = 561$), confirming that the count carries real information about oral status and that the null is not measurement failure.

With both exposures in one model ($n = 561$), self-rated oral health retained its full association (0.71 points per step; 95% CI 0.30–1.11; $p < 0.001$) while tooth loss remained null and, if anything, negative (–0.11; 95% CI –0.27 to 0.06; $p = 0.21$) (Table 4). The association with depressive symptoms is therefore carried by the current, self-rated state and not by the accumulated disease that produced part of that state.

3.5 Pre-registered predictions

Of the four registered predictions, prediction 1 (tooth loss associated, with a gradient) **failed**; prediction 2 (negative control null) held; prediction 3 (interpret as confounding if the control behaved like the exposure) was not triggered; and prediction 4 (tooth loss can reach a causal verdict because its timing is fixed) held procedurally — the temporal criterion was met, but the association it was meant to test does not exist.

4. Discussion

4.1 Principal findings

In a nationally representative sample of US women 1–24 months after a birth, poorer self-rated oral health was strongly and robustly associated with depressive symptoms, while the examiner-counted number of missing permanent teeth was not associated at all — despite predicting self-rated oral health, and despite a sample large enough to exclude associations above roughly 0.09 SD. A pre-registered negative control was null, and the same null for tooth loss appeared in 4,458 women without a recent birth.

4.2 Interpretation

Three explanations were pre-specified for an oral health–depression association: a causal effect of oral disease, a reverse effect of depression on oral state and its reporting, and shared social disadvantage. The contrast between the two exposures discriminates among them.

A causal effect of accumulated oral disease is not supported. Tooth loss is the cumulative, irreversible record of that disease, and it was unrelated to symptoms. If oral disease contributed materially to postpartum depressive symptoms in this population, tooth loss should have shown it.

Reverse causation and shared current state remain plausible. Self-rated oral health is measured at the same moment as the symptoms and is a judgement, not a count. Depression lowers self-rated health across domains, reduces oral self-care, and may make current dental discomfort more salient. The longitudinal finding that depression and stress predict subsequent decline in perinatal oral self-care⁸ fits this direction.

Current oral symptoms not captured by tooth counts are a third possibility consistent with our data: active pain, caries or gingival inflammation could affect mood without yet having produced extractions in women whose median tooth loss is

zero. Our data cannot separate this from perception, since both are concurrent.

4.3 Comparison with other studies

Our null for tooth loss appears to conflict with prospective cohorts in which fewer teeth predicted later depressive symptoms,^{9–11} but the populations differ sharply. Those cohorts studied middle-aged and older adults, with exposure contrasts such as fewer than 10 remaining teeth, or 10–19 versus 20 or more. In our sample of women aged 20–44, median tooth loss was zero and only 3.7% had lost six or more teeth; the range of exposure those studies found informative is largely absent here. Our result therefore bounds the association in a young perinatal population rather than contradicting findings from older adults, where impaired chewing, denture use and visible loss are common.

For self-rated oral health, our estimate extends to the postpartum period associations previously reported during pregnancy^{5,6} and in general adult populations,² and matches the direction reported in the ELSA analysis of oral-health-related quality of life.¹² Mendelian randomization evidence remains contradictory,^{13–16} and one of those analyses found the depression–oral disease relationship disappeared after adjustment for smoking and education,¹⁶ a pattern our socioeconomic and cotinine adjustment also partly reproduced.

4.4 Strengths

The study has four design strengths. First, it uses a nationally representative probability sample with a clinical dental examination, not self-report alone. Second, it exploits an exposure whose timing is fixed by biology, which is rare in cross-sectional data and directly addresses the ordering problem that limits this literature. Third, the negative control was pre-registered, so the null is interpretable rather than merely reassuring. Fourth, the analysis plan, the covariate set, the categories and four numbered predictions were registered before analysis, and the prediction that failed is reported.

4.5 Limitations

1. **Cross-sectional outcome.** Depressive symptoms were measured once. Although tooth loss precedes the birth, we cannot date symptom onset, and a woman with long-standing depression could have accumulated both.
2. **Tooth loss is a coarse proxy for current oral disease.** It does not capture untreated caries, periodontal status, pain or treatment need. A null for tooth loss does not exclude an effect of active oral disease.
3. **Exposure range.** Severe tooth loss is uncommon in women aged 20–44, so precision is limited at the severe end (25 women had six or more missing teeth).
4. **Self-rating and common-method bias.** The positive association may partly reflect a general tendency of depressed respondents to rate health poorly, which no adjustment here can remove.
5. **Unmeasured confounding.** Depression before pregnancy, childhood adversity, sugar intake and dental-care access were not measured. Prior depression is the strongest known risk factor for postpartum depression¹ and could shape both oral state and symptoms.
6. **Screening instrument.** PHQ-9 ≥ 10 identifies probable depression, not diagnosed disorder, and the window spans 1–24 months postpartum rather than the early postpartum weeks.
7. **Missing data.** 14% of the sample lacked a dentition examination and the oral-health rating was not asked in 2007–08; complete-case analysis assumes missingness is unrelated to the outcome given the model.
8. **Multiplicity.** These hypotheses were tested inside a larger search programme; we report false-discovery adjusted values across that programme, but the self-rated finding was identified in the same programme and warrants external replication.

4.6 Implications

For practice. Poor self-rated oral health identifies postpartum women with about three times the screen-positive rate of those rating their oral health good (27.6% vs 8.0%). As a single question already asked in many settings it is a useful flag for psychosocial assessment, and dental visits are a plausible contact point for depression screening. Our results do not, however, support promising improved mental health as a benefit of dental treatment in this population.

For research. The decisive study would measure oral status clinically before or early in pregnancy — including caries, periodontal status and pain, not only tooth count — and depressive symptoms repeatedly afterwards, with pre-pregnancy depression recorded. Randomised trials of postpartum dental treatment could measure mood as a secondary outcome. Until then, the oral health–postpartum depression association should be described as bidirectional or undetermined, not as evidence that oral disease causes depression.

5. Conclusions

Among US women 1–24 months after a birth, poorer self-rated oral health was associated with substantially more depressive symptoms, whereas accumulated tooth loss — which cannot have been caused by depression arising after the birth — showed no association, as did a pre-registered negative control. The association is more consistent with reverse causation or shared current state than with a causal effect of accumulated oral disease.

Figures

Figure 1. Weighted mean PHQ-9 by oral-health exposure

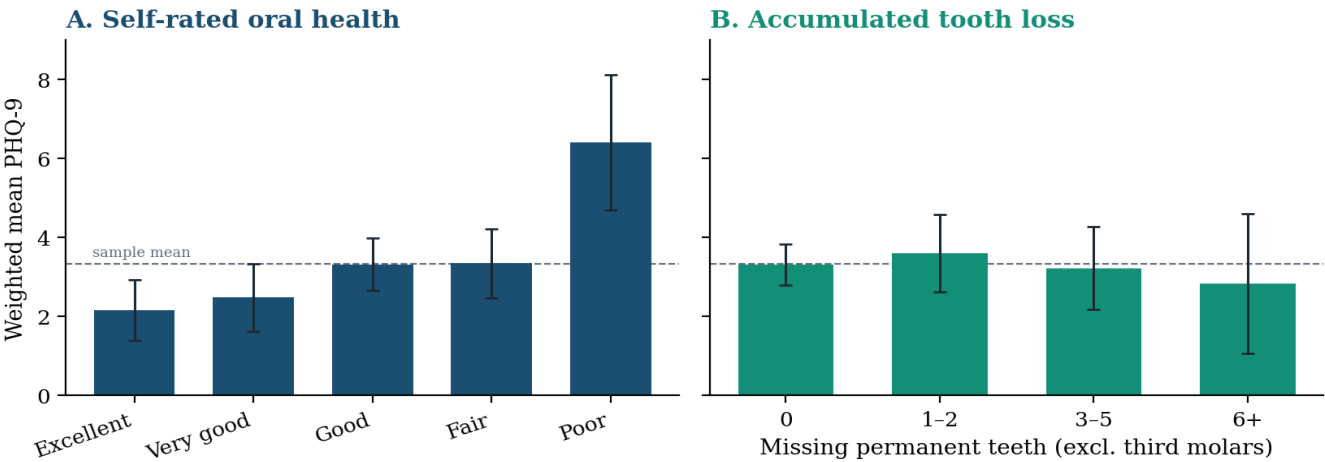


Figure 1. Weighted mean PHQ-9 score with 95% confidence intervals across self-rated oral health (excellent to poor) and across categories of missing permanent teeth (0, 1–2, 3–5, 6+), US women 1–24 months postpartum, NHANES 2007–March 2020. The gradient is monotone for self-rated oral health and absent for tooth loss. Plotted from Table 2. Dashed line marks the sample mean PHQ-9 (3.33).

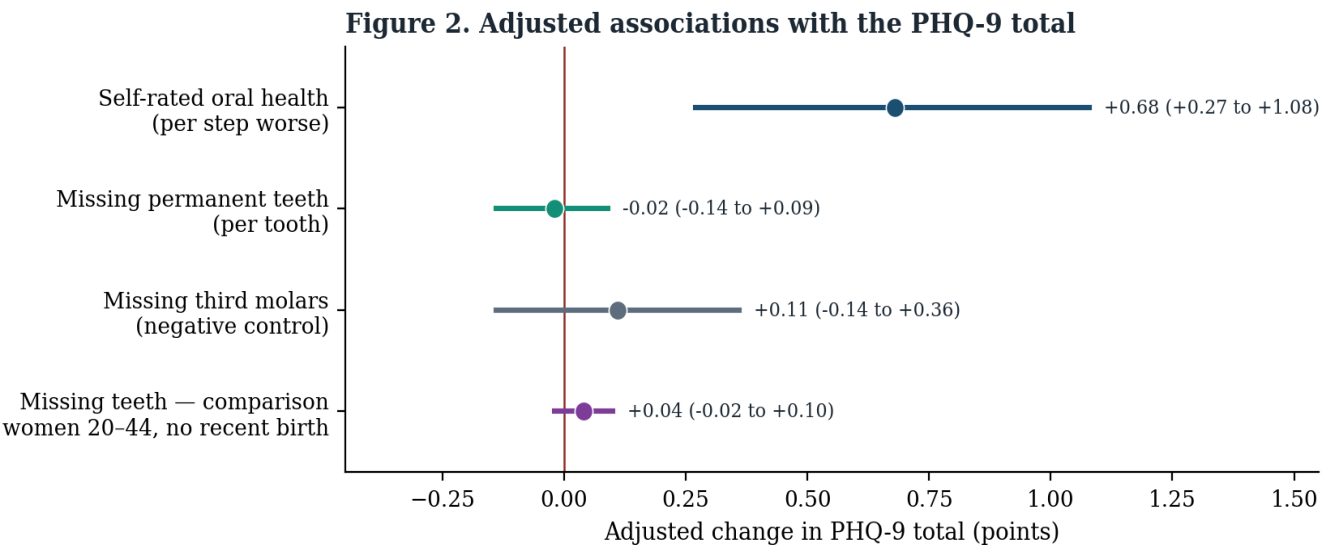


Figure 2. Adjusted associations with the PHQ-9 total for self-rated oral health, missing permanent teeth, missing third molars (negative control), and missing permanent teeth in comparison women aged 20–44 with no recent birth, with 95% confidence intervals on a common scale. Plotted from Table 3. Models adjusted for age (linear and quadratic), race/ethnicity, education, income-to-poverty ratio, nativity, serum cotinine, BMI and survey period.

Tables

Table 1. Characteristics of US women 1–24 months postpartum, NHANES 2007–March 2020

Characteristic	All with PHQ-9 (n = 861)	With dentition examination (n = 742)
Age, years, mean (95% CI)	29.3 (28.8–29.9)	29.7 (29.1–30.3)
Months since birth, mean (95% CI)	11.0 (10.4–11.6)	11.0 (10.3–11.6)
Income-to-poverty ratio, mean (95% CI)	2.27 (2.09–2.44)	2.33 (2.14–2.52)
Body mass index, mean (95% CI)	29.5 (28.7–30.2)	29.5 (28.7–30.3)
PHQ-9 total, mean (95% CI)	3.33 (2.95–3.71)	3.33 (2.92–3.74)
PHQ-9 ≥ 10, %	8.7	8.7
Race/ethnicity, %		
Non-Hispanic White	57.2	58.3
Mexican American	13.9	13.5
Non-Hispanic Black	12.8	12.6
Other Hispanic	8.6	7.8
Other or multiracial	7.5	7.9
Education, %		
Less than 9th grade	3.8	3.4
9th–11th grade	11.5	10.9
High school or GED	23.6	22.7
Some college or associate degree	32.4	32.9
College graduate or above	28.7	30.2
Marital status, %		
Married or living with partner	80.0	80.1
Never married	16.1	15.9
Widowed, divorced or separated	3.9	4.1
Serum cotinine ≥ 3 ng/mL, %	—	23.8
Self-rated oral health, %		
Excellent	8.0	9.0
Very good	18.5	20.3
Good	30.8	33.1
Fair	14.3	15.8
Poor	8.2	9.0
Missing permanent teeth (excl. third molars), %		
0	—	67.2
1–2	—	16.4
3–5	—	12.8
6 or more	—	3.7
Missing teeth, P5 / median / P95	—	0 / 0 / 4
Missing all four third molars, %	—	61.5

Unweighted n; percentages and means weighted to the US population. Cells resting on fewer than five women are suppressed. Percentages for self-rated oral health do not sum to 100 because the item was not asked in 2007–08; cotinine was measured in 710 women.

Table 2. Weighted PHQ-9 by exposure category

Exposure category	n	Mean PHQ-9 (95% CI)	PHQ-9 ≥ 10, % (95% CI)
Self-rated oral health			
Excellent	65	2.15 (1.38–2.92)	suppressed (<5 cases)
Very good	121	2.48 (1.61–3.34)	4.5 (1.7–11.5)
Good	256	3.32 (2.66–3.99)	8.0 (4.6–13.5)
Fair	149	3.35 (2.47–4.22)	8.7 (4.5–16.1)
Poor	60	6.41 (4.70–8.11)	27.6 (16.8–41.9)
Missing permanent teeth			
0	470	3.31 (2.80–3.83)	7.4 (4.9–11.0)
1–2	142	3.60 (2.62–4.58)	13.1 (7.4–22.1)
3–5	105	3.22 (2.17–4.27)	11.4 (5.5–22.1)
6 or more	25	2.83 (1.07–4.60)	suppressed (<5 cases)

Design-based 95% confidence intervals. Cells resting on fewer than five screen-positive cases are suppressed.

Table 3. Adjusted associations with depressive symptoms, women 1–24 months postpartum

Exposure	n	PHQ-9 total, β (95% CI)	p	PHQ-9 ≥ 10, OR (95% CI)	p
Self-rated oral health, per step worse	577	0.68 (0.27 to 1.08)	0.001	1.98 (1.38–2.83)	<0.001
Missing permanent teeth, per tooth	658	−0.02 (−0.14 to 0.09)	0.68	1.01 (0.94–1.09)	0.79
Missing third molars, per tooth (negative control)	658	0.11 (−0.14 to 0.36)	0.39	1.02 (0.85–1.22)	0.82
Missing permanent teeth, per tooth — comparison women	4,458	0.04 (−0.02 to 0.10)	0.23	—	—

Design-based models adjusted for age (linear and quadratic), race/ethnicity, education, income-to-poverty ratio, nativity, serum cotinine, body mass index and survey period; design df = 104. Standardised estimate for missing permanent teeth: −0.015 SD (95% CI −0.087 to 0.057). Robustness values for the self-rated oral health estimate: 0.119 to remove significance, 0.275 to remove the estimate.

Table 4. Both oral exposures in one model (PHQ-9 total, n = 561)

Exposure	β (95% CI)	p
Self-rated oral health, per step worse	0.71 (0.30 to 1.11)	<0.001
Missing permanent teeth, per tooth	−0.11 (−0.27 to 0.06)	0.21

Pathway check, same women: each missing tooth predicted 0.08 (0.03–0.13) worse self-rated oral health, $p < 0.001$.

Declarations

Ethics approval. NHANES protocols were approved by the NCHS Research Ethics Review Board; participants gave written informed consent. Secondary analysis of de-identified public-use data required no additional approval.

Data availability. All data are public. NHANES files and codebooks are available from the Centers for Disease Control and Prevention (<https://www.cdc.gov/nchs/nhanes/>). The National Center for Health Statistics is not responsible for the analyses or interpretations presented here.

Code availability. Analysis code, the pre-registration, the derived-variable definitions and all generated tables are described in Supplementary Files 1–4 of the study repository.

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Competing interests. The author declares no competing interests.

Author contributions. Olaf C. Woodall is the sole author and is responsible for conception, analysis, drafting, revision and approval of the published version.

Use of artificial intelligence. Analyses were executed by a deterministic, version-controlled analysis pipeline. Literature searching, pre-registration text and manuscript drafting were assisted by large language models; all statistical output was produced by the pipeline. The author verified the

content and accepts responsibility for it.

Reporting guideline. STROBE checklist for cross-sectional studies (Supplementary File 4).

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Version. Public working paper, 19 September 2026. Prepared for independent online publication. A journal-formatted variant was drafted with *Community Dentistry and Oral Epidemiology* as a possible eventual venue.

Supplementary files

1. Pre-registration and addenda, including the four numbered predictions.
2. Build and validation report: 177/177 codebook frequency checks reproduced against NCHS published counts; NCHS Data Brief 303 estimates reproduced; CDC published urinary perchlorate geometric means reproduced as a check on subsample weighting.
3. Full model output, adjudication logs and the search ledger for the wider programme, including false-discovery q-values across all 68 hypotheses.
4. STROBE checklist.

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